# Jodh Singh Nahal

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## Education

**University of British Columbia** *Vancouver, BC*
*Bachelor of Applied Science in Computer Engineering* *Expected May 2028*
- Presidential Scholars Award ($60,000), Dean's Honour List, available 4 or 8 months, CGPA: 84.7%.

## Technical Skills

**Languages**: C/C++, ARM Assembly, Python, SystemVerilog, Shell
**Firmware**: Zephyr RTOS, FreeRTOS, CAN/CAN-FD, SocketCAN, SPI, I2C, UART, BLE, JTAG
**Platforms & Tools**: STM32, ESP32, TI C2000, Teensy, GDB, Oscilloscopes, Logic Analyzers, Git

## Experience

**Hardware Verification Engineering Intern** *Vancouver, BC*
*AMD* *Sept 2026 – Present*
- Contributing to verification of the IOHUB subsystem for next-generation AMD processor architectures.
- Verifying IOMMU L1 cache behavior using UVM in a constrained-random testbench.

**Software Test / Embedded Development Co-op** *Burnaby, BC*
*Xantrex* *May 2024 – Aug 2025*
- Developed an object-oriented, multi-threaded, non-blocking C++ firmware architecture on Linux-based Raspberry Pi gateways to parse RV-C CAN frames from STM32 devices and publish structured messages over MQTT.
- Extended firmware support to NMEA 2000 and a proprietary UART battery communication protocol.
- Built Python Hardware-in-the-Loop (HIL) CAN and UART simulators to run 100+ verification tests on the Raspberry Pi gateway, validating edge cases and stress-testing real-time performance.
- Troubleshot CAN/BLE communication faults on embedded inverter and charger platforms using oscilloscopes, logic analyzers, PCAN-USB adapters, and BLE sniffers.
- Achieved 7× device capacity, 90% lower CPU usage, and ∼2 s lower latency after the C++ migration.
- Replaced a ∼60k-line Python implementation with the C++ architecture.

**Firmware / Embedded Systems Developer** *Vancouver, BC*
*UBC Rover* *Sept 2023 – July 2026*
- Developed a 16-channel telemetry system on a TI C2000 (LAUNCHXL-F28069M) using ePWM-triggered ADC interrupts, streaming GPIO/ADC data over SCI/UART.
- Debugged C2000 firmware over JTAG in CCSTUDIO.
- Developed Zephyr RTOS firmware on a Teensy 4.1 for stepper-based control of a 6-axis robotic arm.
- Led migration to a CAN-FD architecture using Moteus r4.11 controllers and BLDC motors for closed-loop torque control and higher bandwidth.
- Automated firmware validation with GitHub Actions CI and Catch2 unit tests to improve regression coverage.
- Led a team of 3 engineers building a distributed embedded control system across ESP32 (FreeRTOS), STM32, Arduino, and TI LaunchPad MCUs over SPI/I2C.

**Undergraduate Research Assistant** *Vancouver, BC*
*UBC Electrical and Computer Engineering — Supervisor: Dr. Lutz Lampe* *May 2026 – Sept 2026*
- Designing embedded power-management PCB hardware in KiCad integrating AC–DC rectifiers, transformers, and switching circuitry.
- Optimizing power extraction of piezoelectric energy harvesting interface circuits (SEH, SSHI, SECE) through LTspice and Simulink/Simscape simulation.

## Projects

**OS161 Kernel Development** | *C, GDB, MIPS, Operating Systems*
- Engineered thread synchronization primitives, including blocking locks and condition variables, to resolve race conditions in kernel space.
- Implemented system calls including fork, execv, and file I/O.

**Memory Allocator Implementation** | *C, Pointers, Systems Programming*
- Developed a custom malloc, free, and realloc using an implicit free list with 8-byte alignment and block headers.